Numerical Analysis Assignments

Fall, 2025

Yiping Cheng
ypcheng@bjtu.edu.cn
15-Sept-2025

School of Electronic and Information Engineering
Beijing Jiaotong University

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Chapter 3 assignment

1. What is the relationship between the multipliers (used to zero out the entries below the pivot entry) in Gaussian elimination for solving $A\boldsymbol{x} = \boldsymbol{b}$ and the entries in L of the LU factorization of A?
2. Find the LU factorization of

$$A = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{bmatrix}$$

   using the standard algorithm provided in my lecture notes.
3. Let $\boldsymbol{b} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{6} \\ \frac{1}{12} \end{bmatrix}$. Solve $L\boldsymbol{y} = \boldsymbol{b}$, where L is obtained in Problem 2.

Chapter 3 assignment (continued)

4. Show that

$$\begin{bmatrix} u_{11} & u_{12} & u_{13} \\ & u_{22} & u_{23} \\ & & u_{33} \end{bmatrix}^{-1} = \begin{bmatrix} u_{11}^{-1} & & \\ & 1 & \\ & & 1 \end{bmatrix} \begin{bmatrix} 1 & -u_{12}u_{22}^{-1} & 0 \\ & u_{22}^{-1} & 0 \\ & & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -u_{13}u_{33}^{-1} \\ & 1 & -u_{23}u_{33}^{-1} \\ & & u_{33}^{-1} \end{bmatrix}.$$

   Then use this fact to solve $U\boldsymbol{x} = \boldsymbol{y}$, where U is obtained in Problem 2 and $\boldsymbol{y}$ is obtained in Problem 3.

5. Write a similar statement about

$$\begin{bmatrix} 1 & & \\ l_{21} & 1 & \\ l_{31} & l_{32} & 1 \end{bmatrix}.$$

   Then use this statement to redo Problem 3.

Chapter 3 assignment (continued)

6. Let $A = \begin{bmatrix} 1 & -4 \\ -2 & 3 \end{bmatrix}$. Compute $\|A\|_1$, $\|A\|_2$, $\|A\|_\infty$.

7. For the linear system $A\boldsymbol{x} = \boldsymbol{b}$ where $A = \begin{bmatrix} 1000 & 0 \\ 0 & 1 \end{bmatrix}$. For the 2-norm, find $\frac{\|\delta \boldsymbol{x}\|}{\|\boldsymbol{x}\|} / \frac{\|\delta \boldsymbol{b}\|}{\|\boldsymbol{b}\|}$ for the following cases ($\epsilon \neq 0$):
   1. $\boldsymbol{b} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\delta \boldsymbol{b} = \begin{bmatrix} \epsilon \\ 0 \end{bmatrix}$
   2. $\boldsymbol{b} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\delta \boldsymbol{b} = \begin{bmatrix} 0 \\ \epsilon \end{bmatrix}$